

Bojan Derajić

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PROFILE

- Research Scientist at AUMOVIO (AI Lab Berlin) and PhD candidate at TU Berlin (IMRC Lab), working on safe learning-based motion planning and control for real-time robotic systems, with multiple patent applications and publications at top-tier robotics venues.

EDUCATION

PhD in Electrical Engineering and Computer Science (Robotics & AI)

2023 - Present

Technical University of Berlin, Germany

- Conduct research on learning-based motion planning and control
- Supervise master's students

MSc in Electrical Engineering and Computer Science (Robotics & Control)

2021 - 2022

University of Belgrade, Serbia

- Thesis: "Adaptive Nonlinear Model Predictive Control for Quadcopters"
- GPA: 10.0/10.0
- Selected courses: Machine Learning, Robotics Systems, Computer Vision, Optimal Control Systems

BSc in Electrical Engineering and Computer Science (Robotics & Control)

2017 - 2021

University of Banja Luka, Bosnia and Herzegovina

- Thesis: "Trajectory Planning" (for manipulators)
- GPA: 10.0/10.0
- Awards: Golden Badge of the University of Banja Luka; Nikola Tesla Medal

EXPERIENCE

AI Research Scientist

04/2023 – Present

AI Lab Berlin, Continental Automotive GmbH / AUMOVIO GmbH

- Develop AI-enhanced motion planning methods (nxtAIM project)
- Optimized real-time performance for mobile robot teleoperation (BeIntelli project)
- Filed 5 patent applications (4 as the first inventor)
- Supervise interns and master's students

Research and Teaching Assistant

12/2021 – 03/2023

Faculty of Electrical Engineering, University of Banja Luka

- Researched the problem of adaptive optimal control for mobile robots
- Delivered tutorials on control systems and AI courses

SELECTED PUBLICATIONS

B. Derajić et al., “CN-CBF: Composite Neural Control Barrier Function for Safe Robot Navigation in Dynamic Environments”, arXiv, 2026 (under review)

B. Derajić et al., “ORN-CBF: Learning Observation-conditioned Residual Neural Control Barrier Functions via Hypernetworks”, ICRA, 2026 (accepted)

B. Derajić et al., “Residual Neural Terminal Constraint for MPC-based Collision Avoidance in Dynamic Environments”, CoRL, 2025

B. Derajić et al., “Learning Maximal Safe Sets Using Hypernetworks for MPC-Based Local Trajectory Planning in Unknown Environments”. IEEE RA-L. 2025

SKILLS

Safety-critical control: Model predictive control (MPC), control barrier functions (CBFs), Hamilton–Jacobi reachability, safety filters, collision avoidance

Learning-based control: Neural CBFs, neural constraints for MPC, observation-conditioned safety filters

Robotics: Ground robots, UAVs, manipulators, hardware deployment, latency optimization, dynamics modeling

Software: Python, C/C++, ROS/ROS2, Gazebo, CasADi, acados, JAX, PyTorch, MLflow, Linux, Git, Docker